Possession

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1 Introduction

The aim of Possession is to produce a fun, addictive, and strategic bullet-heaven game that keeps the player consistently engaged. The intended publisher is Juice Incorporated.

2 Concept

2.1 Mechanics

2.1.1 Main loop

Waves of enemies spawn around the player.
The objective of the player is to survive as long as possible.
The player's score increases by defeating enemies.
The player has the ability to 'possess' enemies they recently killed to use their abilities.
Possessed enemies have an expiration time such that the player will need to continuously swap hosts.
Every few waves, a boss will spawn.

2.1.2 The player

The player's capabilities are intentionally limited to encourage the main mechanic- possession.
The player can shoot bullets that deal percentage based damage so they can quickly find new hosts.
The player will have very low health.
The player's health and the health of the corpses they possess will be separate- the health of the corpse will act as a 'shield' of sorts.

2.1.3 The enemies

All enemies will be unique in some way to force the player to adapt to different play-styles.
Uniqueness of enemies will include stat spreads, weapons, and abilities.
To add variance (and make the game more interesting), certain enemies will have special extended abilities. For example, certain enemies may enhance their attack speed for a certain period, certain enemies may summon allies to assist, certain enemies may deflect projectiles for a period of time, etc.

2.1.4 The menus

Menus include: main menu, restart menu, credits menu, and leaderboard menu
There will be a difficulty 'lever' in the main menu.
Credits to the developers and any assets will be viewable in the credits menu.
The player's top scores/runs will be viewable in the leaderboard menu

2.2 Dynamics

A dynamic that emerges from the possession mechanic is strategical decision making. This is because the expiration time of possessed enemies combined with the lack of health recovery will force the player to make difficult decisions. For example, if the player is currently possessing an enemy with a strong attack, but low remaining health, they may have to decide between taking a risk, sticking with that enemy, or possessing a potentially weaker enemy that has more health.

A dynamic that emerges from the variety of enemies is flexibility and adaption. This is because each enemy will be unique, having different weapons, stat spreads or abilities. Combined with the possession mechanic, the player will need to learn how to counter particular enemies, as well as how to fight as those same enemies, given their capabilities.

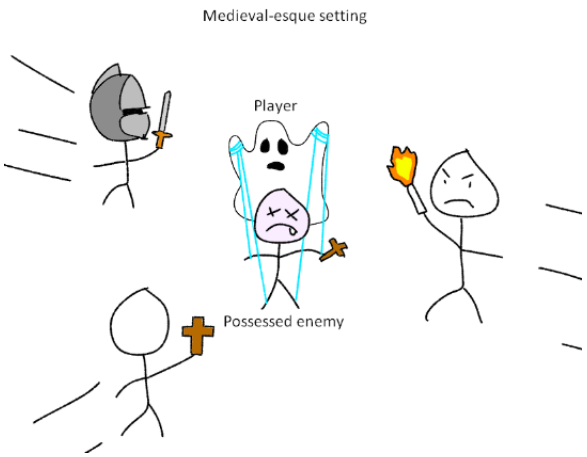


Figure 1: Sketch of the theme

2.3 Aesthetics

2.3.1 Player UI

The view will be top-down and landscape. Sprites will be 2D and simple (having a resolution of 128 by 128).
The UI of the player will be abstracted, only displaying health and ability cool-down.
Individual enemies will not have visible health bars to reduce screen clutter.

2.3.2 Theme/Plot

The player is an amnesiac ghost. They do not remember how or when they died. The only thing they know is that there is something they're supposed to do. Unfortunately for the player, they are being hunted down. Not by demons, not by monsters, but by priests and knights from the holy church.

2.3.3 Player experience

The gameplay mechanics are intended to give the player a diverse experience of different play-styles according to the enemy they 'possess'. Moving between possession targets should be risky and make the player feel vulnerable to emphasize the strategical, decision making aspect.

3 Justification

3.1 Identity

The key factor that distinguishes this game from typical bullet heaven games is progression. The player themselves does not progress in the traditional sense, gradually growing stronger through the gameloop. Instead, the player's strengths and abilities will entirely depend on the enemies they possess. Consequently (intentionally), the player will never become too powerful, and instead, allows players to experience diverse play-styles with each new enemy introduced.

The intended audience for this game fit into the 'achievers' or 'explorers' category. To appeal to the 'achievers' we will implement a scoring system that increases as the player defeats more enemies. We will also implement storage of the top score(s) that players earn. To appeal to the 'explorers', we will have a large variety of enemies for the player to adapt to, learn from, and fight as.

3.2 References

Our game draws inspiration from bullet heaven titles such as 'Vampire survivors' and rogue-like titles such as 'MidBoss'. It is a merge of the concept of stealing enemy skills and top-down, shoot-em-ups.



Taken from:
https://vampire.survivors.wiki/w/Stages
https://wikirby.com/wiki/Inhale

Figure 2: Image of concept

In fact, the game concept could be likened to the idea of a bullet heaven Kirby game- copying opponents' weapons/abilities to use against them. Except, the abilities do not last indefinitely (doing so would encourage players to only possess enemies that they deem the strongest instead of hopping between hosts).

3.3 JUICE INC

The game will implement a time based wave system to spawn enemies at fixed intervals rather than waiting for all existing enemies to die. This applies pressure to the player as they will need to constantly kill enemies as fast as possible. If players try to stall or avoid enemies, they will be punished with increasing waves of enemies. This encourages the player to make act proactively rather than passively so that they don't get overwhelmed by the enemies.

However, some players will be able to kill all enemies of a wave much faster than others. In this case, the next wave of enemies will spawn before the fixed interval timer, allowing the player to beat waves and the game quickly. This will encourage competition between players that want to beat the game as fast as possible, while casual or new players can play at the intended difficulty or pace of the game. A way to further reward players that beat waves quickly is to provide them additional score for millisecond of time they save in each wave. This ensures that the score system doesn't become obsolete when the game gets heavily optimised by certain players.

3.4 Replay-ability

Enemies are randomized in each run to force the player to make different decisions; the strategy that worked for a previous run may not perform as well in the next run. With each run, the player will always learn something new- whether that is how to counter a particular enemy, or how to maximize survival while possessing a particular enemy, the randomized environment will benefit replay-ability.

4 Planning

Tasks will be compiled into a 'TODO' list and evenly divided between the group.
The project itself will be stored on github to allow for collaborative work.
Tasks completed will be checked and confirmed by at least two other members on the team before moving to the next task to ensure quality and functionality.
Particularly large or creative tasks, such as the design of bosses, will be designed in collaboration.

4.1 Prototype

Expected completion date- 29/10/2026
Player success criteria:

- The player can move in the four cardinal directions.
- The player has a basic attack that deals damage to enemies.
- The player can possess enemies they kill.
- The UI clearly indicates the health of the player (and any enemy they are possessing)

Enemy success criteria:

- At least 3 enemy variants.
- Enemies are deleted after a short timer upon dying if they are not possessed.
- Enemies act autonomously when not possessed (move toward and attack the player).

Game loop success criteria:

- Enemies are spawned in waves.
- The spawning of said waves is stagnated by a timer.

4.2 Milestone 1- Enemy variants

Expected completion date- 15/11/2026
Justification- Four person team. This means 4 normal enemies and 1 'special' enemy each. The bosses receive special consideration and be developed together. An 'enemy' superclass will be created in which all the enemies can inherit from to simplify the development process. Due to the isolation of enemy development, it will not create git conflicts.
Success criteria:

- Enemy variants include 'melee' and 'ranged'.
- Enemies are all unique in some way.
- At least 16 enemy variants
- At least 4 'special' enemies with extended abilities (as outlined in the enemies section of the concept).
- At least 2 boss enemies (not all bosses may have a special gimmick, hence it is separate from the 'special' enemies criteria).

4.3 Milestone 2- Game loop

Expected completion date- 29/11/2026
Justification- With the enemies designed, the final component of the game will be putting them together (in waves) and ensuring that they work as intended and are balanced. Creating waves of enemies would be hardly possible without the enemies- which is why they were prioritised.
Success criteria:

- Difficulty should ramp up with each wave.
- At every nth wave, a boss should spawn.
- A 'difficulty' lever that allows the player to select between 'easy' 'normal' or 'hard'. Said lever should affect the base stats of enemies and the player.
- The score of the player should be recorded, such that the player is able to view their top score(s) when needed.
- There is a 'game over' screen upon death that allows restarting.
- There is a main menu that allows the player to start playing.

4.4 Milestone 3- Balancing, debugging and testing

Expected completion date- 05/12/2026
Justification- It's important to get non-biased feedback when it comes to games- as the developers are more likely to be good at their game than someone who has never seen it before. The reason this milestone is later than the others is because the actual game needs to be in some working condition before people can play it.
Success criteria:

- The game loop should be fun and engaging- random volunteers from the 'esports' soceity will be asked to test the game, their feedback taken into consideration. A main focus on getting their input regarding:
    1. Balancing
    2. Aesthetics
    3. Their strategy when playing
    4. Any improvements they suggest
- The 'possession' mechanic should be meaningful. The relevant dynamics as described in the previous section should be apparent.
- No enemy should feel unbalanced- this would undermine the 'possession' mechanic.
- The player should not feel that the game is unfair or unreasonable.
- The difficulty 'lever' should have a noticeable impact during gameplay.

4.5 Milestone 4- Assets

Justification- The assets include sprites and SFX. It is of least priority for this game as the focus is not on appearances. Expected completion date- 12/12/2026
Success criteria:

- All the enemies in the game should have relevant sprites according to the theme.
- All the UI should have sprites.
- The player should have a relevant sprite according to the theme.
- There should be background music in any and all menus.
- There should be background music during gameplay.
- Relevant sound effects should play during certain actions.
- There should be a way to view credits for bgm, sfx, assets, and developers.